

Test Execution Report

OldPhonePad Coding Challenge — xUnit / .NET 8

Project: OldPhonePad Coding Challenge
Executed By: Mauricio Batista
Execution Date: 2026-06-01
Framework: xUnit .NET 8
Execution Time: ~2.7 s
Total Tests: 27
Passed: 27
Failed: 0
Overall Result: PASS

1. Summary

All 27 test cases executed successfully against the OldPhonePad method on 2026-06-01. The test suite covers PDF examples, single key presses, space separators, cycling wrap-around, backspace behaviour, key 1 special characters, all-key validation, and defensive/edge-case scenarios. No failures or errors were recorded.

2. Execution Results

Test ID	Executed By	Execution Date	Result	Actual Output	Notes
TC-01	Mauricio Batista	2026-06-01	Pass	E	-
TC-02	Mauricio Batista	2026-06-01	Pass	B	-
TC-03	Mauricio Batista	2026-06-01	Pass	HELLO	-
TC-04	Mauricio Batista	2026-06-01	Pass	TURING	-
TC-05	Mauricio Batista	2026-06-01	Pass	A	-
TC-06	Mauricio Batista	2026-06-01	Pass	D	-
TC-07	Mauricio Batista	2026-06-01	Pass	W	-
TC-08	Mauricio Batista	2026-06-01	Pass	AA	-

Test ID	Executed By	Execution Date	Result	Actual Output	Notes
TC-09	Mauricio Batista	2026-06-01	Pass	A B	-
TC-10	Mauricio Batista	2026-06-01	Pass	S	Wrap-around: 4 presses on key 7 cycles back to S (PQRS)
TC-11	Mauricio Batista	2026-06-01	Pass	P	Wrap-around: 5 presses on key 7 cycles back to P (PQRS)
TC-12	Mauricio Batista	2026-06-01	Pass	A	Wrap-around: 4 presses on key 2 cycles back to A (ABC)
TC-13	Mauricio Batista	2026-06-01	Pass	B	-
TC-14	Mauricio Batista	2026-06-01	Pass	""	Backspace on empty buffer is silently ignored
TC-15	Mauricio Batista	2026-06-01	Pass	""	-
TC-16	Mauricio Batista	2026-06-01	Pass	""	-
TC-17	Mauricio Batista	2026-06-01	Pass	BA	-
TC-18	Mauricio Batista	2026-06-01	Pass	AA	-
TC-19	Mauricio Batista	2026-06-01	Pass	A	Excess backspaces stop at empty; subsequent typing resumes normally
TC-20	Mauricio Batista	2026-06-01	Pass	&	-
TC-21	Mauricio Batista	2026-06-01	Pass	'	-
TC-22	Mauricio Batista	2026-06-01	Pass	(	-
TC-23	Mauricio Batista	2026-06-01	Pass	ADGJMPTW &	-
TC-24	Mauricio Batista	2026-06-01	Pass	ArgumentNullException thrown	Initial run failed. See BUG-01, re-executed after fix
TC-25	Mauricio Batista	2026-06-01	Pass	""	-
TC-26	Mauricio Batista	2026-06-01	Pass	""	-
TC-27	Mauricio Batista	2026-06-01	Pass	""	-

3. Defects Found During Testing

Defect ID	Test Case	Description	Status
BUG-01	TC-24	Method threw NullReferenceException instead of ArgumentNullException on null input.	Fixed

Defect ID	Test Case	Description	Status
		Fix applied: Added <code>ArgumentNullException.ThrowIfNull(input)</code> as guard clause at the top of the method.	

4. Quality Metrics

Metric	Type	Formula & Calculation	Result	Interpretation
Test Effectiveness	Process	Defects in test / (Defects in test + Defects post-release) x 100 $1 / (1 + 0) \times 100 = 100\%$	100%	All defects caught before shipping. No post-release escapes.
Test Execution Coverage	Process	Tests executed / Tests planned x 100 $27 / 27 \times 100 = 100\%$	100%	Every planned test case was executed. Full coverage achieved.
Error Fix Rate	Product	Defects fixed / Defects found x 100 $1 / 1 \times 100 = 100\%$	100%	BUG-01 was identified and fixed before release. Zero open defects.
Defect Density	Product	Defects found / Methods in codebase $1 / 3$ methods (OldPhonePad, CommitKey, ThrowIfNull) = 0.33 defects/method	0.33	Acceptable for a small codebase. The single defect was found in OldPhonePad.
Defect Leakage	Product	Defects post-release / Total defects found x 100 $0 / 1 \times 100 = 0\%$	0%	Zero defects escaped to end users. Testing contained all issues within the cycle.